Alexandra Ding

Cambridge, MA | (650) 660-6688 | alexding@mit.edu | junypr.github.io

EDUCATION

Massachusetts Institute of Technology **Cambridge, MA**

B.S. in Computer Science and Engineering September 2024 – May 2028 (expected)

GPA: 5.0

Relevant Coursework: Interpreters, Efficient AI, Dynamics and Controls, Software Construction, Computer Systems, Algorithms II, C and Assembly, Processor Design, Linear Algebra, Differential Equations

SKILLS

- **Programming:** Python (OpenCV, multiprocessing, matplotlib, pytest, pandas, PyTorch), Typescript, C++, ROS, C, CSS, HTML, Java, Bash, Minispec, RISC-V assembly, Git, Linux
- **Analysis & CAD:** Statics, FEA, CFD (SimScale, ParaView), XFLR5**,** Fusion 360, Onshape, KiCAD

WORK EXPERIENCE

Relativity Space **Long Beach, CA**

Robotics Software Engineer Intern May 2026 – August 2026

- Architected a real-time rigid body tracking pipeline in C++, designing the multi-threading model and ROS orchestrator to ensure performance at 60 hz
- Developed Python validation tools to verify the spatial accuracy of a multi-camera photogrammetry rig, proving sub-millimeter precision and system reliability over multi-hour continuous operations

MIT Materials Research Laboratory **Cambridge, MA**

Student Worker - AI Tools Development January 2026 – present

- Architected an automated data pipeline utilizing web scrapers and APIs to ingest data from thousands of academic papers, job postings, and patents into an optimized SQLite3 database
- Engineered a semantic matching engine using LLM-driven keyword extraction and vector embeddings
- Operationalized the platform to rapidly match inbound corporate technical queries with relevant researchers and generate targeted outreach leads for MIT faculty

Taşan Group **Cambridge, MA**

Undergraduate researcher September 2025 – December 2025

- Fine-tuned Mistral 7B Instruct v0.3 to automate initial feedback on scientific drafts, reducing PI overhead
- Deployed LLM locally using Hugging Face tools to maintain data privacy for unpublished research

NASA Ames Unitary Plan Wind Tunnel **Mountain View, CA**

Intern June 2025 – August 2025

- Characterized the uncertainty of a computer-vision-based angle of attack measurement system using Monte Carlo
- Developed and optimized a Python pipeline using OpenCV and multiprocessing to separately quantify error from detection and from calibration across thousands of large-format images

MIT Sea Grant **Cambridge, MA**

Undergraduate researcher October 2024 – May 2025

- Designed, fabricated, and assembled custom PCBs for an autonomous oyster-basket flipping robotic boat, integrating a Raspberry Pi, Pixhawk flight controller, and I^2C-to-PWM breakout boards

EXTRACURRICULAR ACTIVITIES

Design Build Fly **Cambridge, MA**

Aerodynamics subteam lead September 2024 – May 2026

- Designed and CADed wing for 2026 competition plane, creating a parametric, reusable design and emphasizing manufacturability
- Sized tail using XFLR5 to ensure static and dynamic stability
- Led wing manufacturing, including laser cutting, composite layups, and Monokote application